Experiment 7

API Integration with Zoho Creator

Aim

To integrate Zoho Creator with external applications and services using Application Programming Interfaces (APIs) to enable secure data exchange, automate business processes and extend the functionality of the Transport Management System.

Theory

API integration allows applications to communicate and exchange data with external systems, enabling developers to retrieve, create, update and manage records between applications. In the Transport Management System, APIs are used to read vehicle and trip data into Zoho Analytics or external systems and to call external services from Deluge using connections and **invokeUrl**.

Procedure

1. Log in to Zoho Creator and open the Transport Management System application.
2. Identify the external application or service to integrate (Zoho Analytics for reporting, and the Creator REST API for data exchange).
3. Configure the required API access and authentication settings (OAuth connection in Zoho Creator).
4. Generate or configure the necessary API credentials and scopes.
5. Use supported integration options or Deluge functions to send and receive data.
6. Configure the request URL, HTTP method, headers and required parameters.
7. Test the integration and verify that data is exchanged correctly.
8. Handle errors and validate responses to ensure reliable operation.

Applications of API Integration in this Project

- Sending Vehicle_Details and Trip_Scheduling data to Zoho Analytics for advanced charts.
- Sending automated email, SMS or WhatsApp notifications when trip status changes (future enhancement).
- Retrieving live vehicle location from an external GPS tracking service (future enhancement).
- Retrieving route distance and travel time from a maps service for Route_Details.
- Synchronizing fuel and maintenance records with an external fleet system (future enhancement).

Request Details

Item	Value
Method	GET
Target	All_Trips_List report through the Creator REST API
Authentication	OAuth connection created in Zoho Creator (read scope on the report)
Used for	Reading trips and vehicles for reports, dashboards and external systems

Sample Deluge: Fetching Records Through a Connection

Replace the placeholders with your own account owner name and connection link name.

```
baseUrl = "https://creator.zoho.in/api/v2/<account_owner>/";
response = invokeUrl
[
    url :baseUrl + "transport-management-system/report/All_Trips_List"
    type :GET
    connection :"creator_connection"
];
info response;
if(response.get("code") == 3000)
{
    records = response.get("data");
    info "Records fetched: " + records.size();
}
else
{
    info "API error: " + response;
}
```

Process Diagram

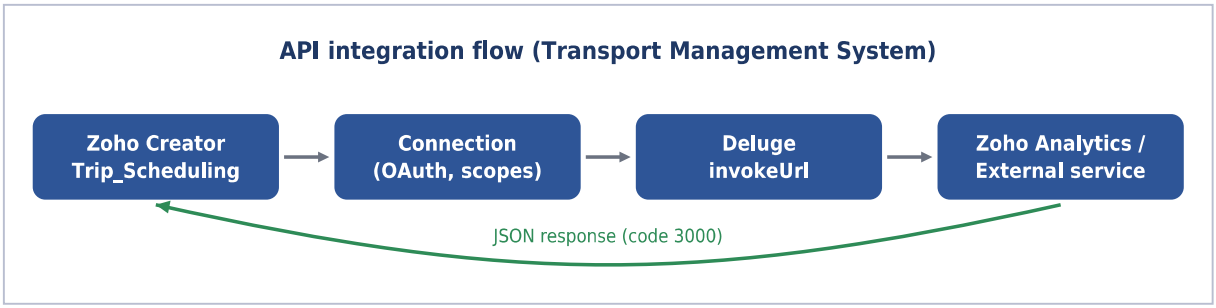


Figure 1: Diagram of the API integration flow (illustrative diagram, not a screenshot).

Expected Response Format (Illustrative)

```
{
  "code": 3000,
  "data": [
    {
      "Trip_ID": "TRP-001",
      "Vehicle": { "ID": "<record_id>", "display_value": "TN 01 AB 1234" },
      "Driver": { "ID": "<record_id>", "display_value": "Arun Kumar" },
      "Route": { "ID": "<record_id>", "display_value": "Chennai - Vellore" },
      "Trip_Status": "In Progress"
    }
  ]
}
```

Error Handling Checklist

- Check that the response code is 3000 before using the data.
- Log the full response when the code is different and notify the administrator.
- Keep credentials in the connection; never write tokens inside the script.
- Fetch only the required records using criteria to avoid slow calls.

Metrics Built in Zoho Analytics from the Sample Data

Sample data for this lab record. Values shown are the expected output for these sample records.

Metric	Value
Total trips	5
Average route distance	114.8 KM
Trips by status	Scheduled 1, In Progress 1, Completed 2, Cancelled 1
Trips by route	Chennai – Vellore 2, Chennai – Kanchipuram 2, Chennai – Pondicherry 1
Vehicles by status	Available 3, On Trip 1, Under Maintenance 1

Result

The integration of Zoho Creator with Zoho Analytics and the Creator REST API for the Transport Management System was designed, and the request, response and error-handling approach was documented; data exchange is to be verified using the steps above.

Conclusion

API integration in Zoho Creator enables communication between applications and external services. It supports data exchange, workflow automation and improved operational efficiency. By configuring APIs with appropriate authentication, validation and error handling, developers can build more flexible and connected applications.